



`model-card:ncd-asthma:modular:v1`

Asthma Modular Composition

This card describes the current asthma model structure used for the CR1 run.

Identity

Composition record	<code>composition:ncd-asthma:modular:asthma-cr1:v1</code>
Subject	<code>ncd-asthma:modular</code>
Runtime	2025-2046; synchronous iterations 10
Signature	72398fbfee606f736b64fc4f79e3cd9e1d9f11068ac9596db928cfe66a275525
Components	

Component	Role	Contract
Demographic substrate	<code>compiled_together</code>	Current asthma-local lowering of the demographic substrate. Desired external owner is the existing botech demographic model.
Asthma epidemiology core	<code>compiled_together</code>	Asthma state machine, disease incidence, prevalence allocation, remission, asthma mortality, and risk-modified incidence surrogate.
Tobacco risk factor	<code>link_then_inject_bundle</code>	Tobacco exposure, intervention effects, relative risk, PAF calculation, and the resolved incidence modifier injected into asthma epidemiology.
Low-dose beclometasone intervention	<code>compiled_together</code>	
High-dose beclometasone intervention	<code>compiled_together</code>	
Oral prednisolone intervention	<code>compiled_together</code>	
Inhaled short-acting beta agonist intervention	<code>compiled_together</code>	
Burden metrics	<code>downstream_handoff</code>	
Economic metrics	<code>downstream_handoff</code>	
Resource handoffs	<code>downstream_handoff</code>	

Use and limits

The asthma model runs asthma epidemiology, selected clinical interventions, tobacco risk-factor effects, and downstream burden, resource, and economic event streams.

The modular composition is currently rejoined into one executable model for the Rust runtime. The run certificate records what was run; it does not encode future comparison intent.